

Mahdi Salem

Lead Data Engineer · Power BI Architect · Operational Intelligence Leader

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Turning operational complexity into decision intelligence.

I don't build dashboards for people to look at — I build them for people to see with. As a Lead Data Engineer and Power BI Developer, I transform fragmented operational data into trusted decision systems that reveal patterns, expose risks, clarify interdependencies, and guide measurable action across production, quality, and executive operations — on demand, without added overhead. Over a decade of hands-on experience designing enterprise data architectures, ETL/ELT pipelines, and semantic models has gone into deploying 150+ Power BI and PowerApps solutions across aerospace and manufacturing environments. I lead and mentor a cross-functional team of developers and data engineers, while drawing on a broader background in organizational development, change management, and executive-level strategy to keep the data platform tightly connected to the business decisions it needs to serve. Comfortable operating at every level — from writing DAX and optimizing a semantic model to leading a team, redesigning a business process, or briefing an executive sponsor.

Highlights of Professional Qualifications

- Microsoft Certified across Power BI, PowerApps, Azure, and Excel; experienced designing scalable data architectures on Azure, Databricks, and Microsoft Fabric.
- 10+ years of agile design, development, and deployment of BI solutions — Power BI dashboards, semantic models, and ETL/data-integration pipelines.
- 10+ years of consultative business analysis, engaging stakeholders to build data flows and logical/physical models (Kimball, Data Vault, 3NF) and architectural frameworks for analytics and reporting.
- 10+ years applying statistical and analytical methods to complex business questions over large datasets, with attention to data integrity, security (RLS, OLS, data masking), and compliance.
- Led reporting frameworks and enterprise analytics platforms incorporating metadata/master-data management and reusable KPIs, calculations, and DAX measures.
- Resolved data discrepancies and wrote specifications across Data Warehouse, MDM, ETL, and BI domains, with a focus on scalability and performance.
- Built ad-hoc analysis and add-ons using PowerApps, TypeScript, VBA, DAX, and Python (e.g., openpyxl); comfortable troubleshooting semantic model refresh and performance issues.

Professional Experience — Data & BI Leadership

Lead Data Engineer & Power BI Developer

Aug 2018 – Present

Bombardier Aviation

Mississauga, ON · Remote

- Designed, developed, and maintained enterprise-grade SSIS packages across a multi-factory, multi-country footprint, improving data-flow accuracy and pipeline efficiency between core operational systems.
- Engineered and deployed over 150 Power BI reports, dashboards, and PowerApps solutions supporting Production Andon teams, Quality Assurance, and PMO governance, including 10+ standalone PowerApps built for fast, low-code process automation.
- Initiated and implemented a Data Quality KPI framework reconciling SAP, MES, and local production databases, materially improving shop-floor data integrity.
- Integrated Azure Data Lake and Azure Portal services to monitor and optimize Power BI Premium capacity usage, reducing resource overhead while maintaining query performance.
- Designed and launched an adaptive OSW (Outstanding Work) Production Tracking web application (.NET / Angular) to structure highly volatile, non-standard assembly tasks.
- Engineered a web-based attendance system with an administrator/supervisor dashboard using Python, Django, JavaScript, jQuery, and SQL for the OSW team.
- Created standardized report templates and mock-up frameworks under the company-wide Bombardier Operational Excellence (BOE) initiative, establishing best practices adopted across future reporting rollouts.
- Built 10+ rapid-response VBA applications automating cross-departmental communication and eliminating manual data entry.
- Led a team of up to 4 developers and data engineers through detailed planning and role clarity; coached and mentored team members, tailoring guidance and development opportunities to help each person contribute their best work.

Senior Data and Process Analyst

Jul 2018 – Aug 2018

Triumph Group (Contract)

Greater Toronto Area, ON

- Engineered a custom VBA application to automate Crew Load Chart generation, cutting processing time from roughly 3 hours to 2 minutes and eliminating manual entry errors.
- Built a Python-based web application for daily attendance management with integrated Crew Load analytics to measure shift efficiency and optimize labor allocation.
- Architected a multi-tiered reporting framework covering daily, weekly, and monthly operational metrics at both macro and aircraft-structural levels.
- Developed real-time alerting mechanisms and dashboards delivering threshold notifications to line and executive managers to prevent project delays.

Process Improvement Lead

Dec 2017 – Jun 2018

AAA Canada — Aerospace & Transportation

Greater Toronto Area, ON

- Built a Manpower Mapping application distinguishing touch vs. non-touch labor, improving shop-floor visibility and reducing non-value-added idle time.
- Architected a Line Logistic Cage tracking system giving real-time status on job kits and workability post-punch-out, preventing downstream assembly bottlenecks.
- Developed an automated Rapid Non-Conformance (RNC) reporting tool that cut generation time to 3 minutes, accelerating quality-containment response.
- Digitized Daily Management Meeting (DMM) reporting into an automated dashboard, lifting reporting efficiency by 200% while removing manual data entry.

Senior Logistics Lead

Aug 2016 – Dec 2017

AAA Canada — Aerospace & Transportation

Greater Toronto Area, ON

- Engineered custom kitting-tracking software and workflows giving full kit-status visibility for non-standard OSW tasks across Rear Fuselage and Cockpit assemblies.
- Architected a centralized database consolidating scope-of-work requirements, over-issue requests, and RNC/SNC non-conformance records into a single source of truth across regions.
- Managed end-to-end SAP inventory for parts, tooling, and hardware, proactively identifying shortages and following up with vendors to prevent FAL line stoppages.
- Developed periodic shortage and hardware-delivery forecasting reports giving top management clear visibility into material risk.

Logistics Lead

Aug 2015 – Jul 2016

Anjaz Group (Contract)

Greater Toronto Area, ON

- Ran strategic procurement negotiations that reduced costs and secured favorable vendor payment terms across a competitive supplier base.
- Managed end-to-end vendor relationships, resolving shipment, quality, and contract-compliance issues, and built proactive contingency plans for critical equipment.
- Served as primary operational liaison for Authorized Service Provider (ASP) networks, and built a closed-loop customer-quality-assurance intake and root-cause process.

Leadership & Consulting Background

Director, Systems and Methods Development / Business Technology & Excellence

2013 – 2015

Persia Oil and Gas Drilling Co. of Qeshm — Tehran, Iran

- Built the company's first enterprise process map and documentation framework, enabling its inaugural drilling project to launch in 3 months, and delivered an audit-verified 15% productivity gain.

VP, Marketing & International Affairs / Manager, Systems & Method Engineering
Oil Exploration Operation Company — Tehran, Iran

2011 – 2013

- Promoted to VP after repositioning Marketing into the company's core bidding engine and leading 31 process-improvement projects in the first 30 days of the prior role.

Lead Business Consultant

2001 – 2012

Independent — Various Companies, Tehran, Iran

- Directed 30+ management-consulting engagements in Strategic Planning, Business Process Re-engineering, TQM, and Six Sigma, authoring 100+ winning proposals and RFQs.

Technical Expertise

BI & Analytics	Power BI (Direct Lake, Copilot, semantic models), DAX, Power Query (M), Tableau, SAS, Cognos
Data Engineering	Microsoft Fabric (OneLake, Lakehouse, Data Factory, Real-Time Intelligence), Databricks (Delta Lake, Unity Catalog, PySpark), Azure Synapse, SSIS, ETL/ELT & medallion architecture
Databases	SQL Server, MySQL, Oracle (10g/12c), Azure Cosmos DB; relational and semi-structured (JSON/API) data
Backend	Python (Django, Flask, Pandas, NumPy, SciPy, scikit-learn, Matplotlib, openpyxl), PySpark, KQL, R, Node.js/Express, ASP.NET Core
Frontend	React, Angular, TypeScript, Bootstrap, CSS, jQuery, D3.js, Lodash, Moment.js
Low-Code / No-Code	PowerApps, Power Automate, WordPress, Joomla, Drupal
Governance & Tooling	Microsoft Purview, RLS/OLS, Git, GitHub, Bash, PowerShell, Unix/Linux

Education & Training

Master of Business Management, Marketing
Allameh Tabatabaie University, 1996 – 1999

Bachelor's Degree, Insurance Management

PROFESSIONAL TRAINING

Business Analytics
Duke University, 2016 – 2017

CS50's Web Programming with Python and JavaScript
Harvard University, 2020

Microsoft Power Platform (Power BI, PowerApps, Power Automate)
2020 – 2021

Certifications

Microsoft Certified: Power BI Data Analyst Associate

Microsoft Certified: Data Analyst Associate

Microsoft Certified: Power Platform App Maker Associate

Microsoft Certified: Power Platform Fundamentals

Microsoft Certified: Azure Data Fundamentals

Microsoft Office Specialist: Excel 2016 Expert

Six Microsoft badges held in total; some pending renewal.